PHYS7101: NUMERICAL METHODS AND DATA ANALYSIS · LECTURE 5

How sure are we?

Describing data, propagating uncertainty
and testing hypotheses

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Where we left off

From Lectures 3 and 4

- Fitted parameters come with uncertainties and correlations
- χ^2/ν and residuals judge a fit
- Independent errors add in quadrature; offsets add linearly
- Unmodelled time variability is not noise

Today

- Describing data: location, spread, distributions
- Propagating uncertainty into computed quantities
- Testing hypotheses: is a difference real?
- One running example: measuring g with a pendulum

In every step the key question will be the same: are the values independent?

Three questions about any result

Task	Question	Tools	Part
Describe	What do the data look like?	Plot in order; median, quantiles; a distribution family only when known	A
Propagate	How uncertain is a quantity computed from them?	Bootstrap; Monte Carlo; the linear rule as a shortcut	B
Test	Is an observed difference real, or chance?	Permutation and rank tests; parametric tests when the distribution is known	C

The order matters

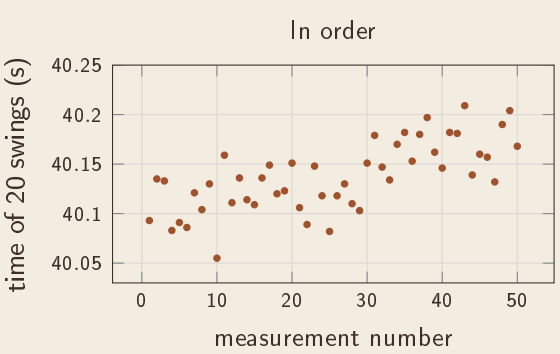
A wrong description (e.g. missed drift or a bad value) gives a wrong uncertainty in Part B and a wrong test in Part C.

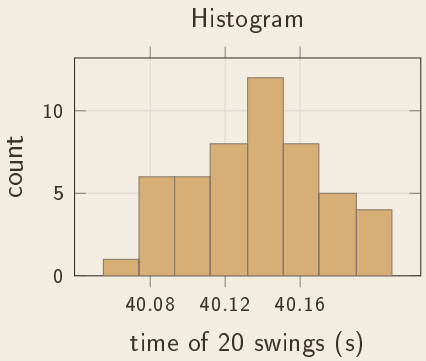
PART A

Describing data

What do the values look like – in order, and as a distribution?

Plot the values in the order they were taken





The histogram looks like ordinary scatter. In order, the times drift upwards: the first 25 average 40.115 s, the last 25 average 40.159 s. Successive values are not **independent** (independent: knowing one value tells nothing about the next).

Location and spread

Values $x_1, \ldots, x_N$; here a new set of $N = 9$ timings of 20 swings: 40.12, 40.18, 40.09, 40.15, 40.11, 40.21, 40.13, 40.16, 40.10 s. **Median**: the middle value when sorted (even N: the mean of the two middle ones).

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i, \qquad s = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2}.$$

median	40.130 s
mean $\bar{x}$	40.139 s
standard deviation s	0.040 s

s describes the spread of **single** measurements: here 6 of the 9 timings lie within $\bar{x} \pm s$.

Why $N - 1$?

Deviations are measured from $\bar{x}$, fitted to the same data, so one value is used up: $\nu = N - 1$ degrees of freedom (as for χ^2, Lecture 3).

One bad value: mean versus median

	9 timings	+ one value of 41.14 s
mean	40.139	40.239
standard deviation s	0.040	0.319
median	40.130	40.140
MAD (median absolute deviation)	0.044	0.052

MAD, a robust spread: take the median; take each value's distance from it, $|x_i - \text{median}|$; take the median of these distances; multiply by 1.4826 so that for Gaussian data it estimates the same σ as s. The median and MAD hardly notice the bad value; the mean and s do.

Never delete a value silently

Find the cause (here: the watch stopped one second late), and report what you did. A value that is merely surprising may be the physics.

Nonparametric and parametric descriptions

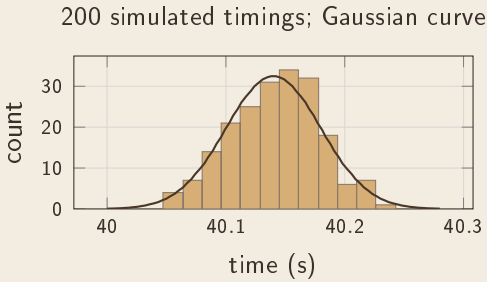
	Nonparametric	Parametric
Assumes	no shape; uses order, ranks, quantiles*	a family (Gaussian, Poisson)
Centre and spread	median, MAD; also $\bar{x}$, s (no shape implied)	$\bar{x}$, s as estimates of μ, σ
Strength	any shape; median, MAD robust to bad values	short formulas; uses a known shape
Risk	ignores a shape that physics gives (a count N: $\sigma = \sqrt{N}$)	wrong if the shape is wrong; few values cannot check it

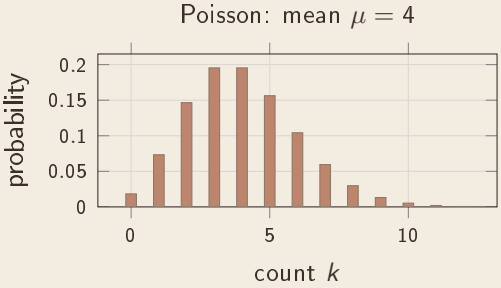
*rank: position in the sorted list; quantile: the value below which a given fraction lies (median: 50%).

Which is steadier depends on the data, not on the family

Simulated: for Gaussian data the median varies from sample to sample 1.2 times more than the mean; with bad values or heavy tails (frequent large deviations), 2–3 times less. Parametric is justified only if the distribution is known from outside the data.

When the distribution is known: Gaussian and Poisson





Gaussian arises when many small independent effects add up. 68.3%, 95.4%, 99.7% of values lie within $\pm 1, 2, 3$ standard deviations of the mean.

Poisson counts of independent events, $P(k) = \mu^k e^{-\mu}/k!$; standard deviation $\sqrt{\mu}$ (the counting data of Lecture 3).

Use only when physics or an independent calibration supplies the family; check it against the data (histogram, plot in order).

Spread of values versus uncertainty of the mean

How far is the mean of N values likely to be from the value it estimates? If the values are **independent**, the standard error of the mean is

$$\boxed{s_{\bar{x}} = \frac{s}{\sqrt{N}}}$$

9 timings: $0.040/\sqrt{9} = 0.013$ s.

N	s	$s/\sqrt{N}$
10	0.04	0.013
100	0.04	0.004
1000	0.04	0.0013

If successive values are correlated

Simulated: 1000 samples of a slow oscillation (amplitude 0.5) plus noise – random values independent from sample to sample (Gaussian, $\sigma = 1$). $s/\sqrt{N} = 0.034$, but over 4000 such records the mean scatters by 0.056 – **1.7 times more**. Fewer values are effectively independent.

Quoting a result

Write

$T = 2.0069\,\mathrm{s}$, 68% interval 2.0063–2.0076 s; symmetric, so also $T = 2.0069 \pm 0.0007\,\mathrm{s}$

same decimal place throughout; 1–2 significant figures in the uncertainty; units.

Not

$T = 2.006944 \pm 0.000660\,\mathrm{s}$

digits beyond the uncertainty carry no information.

The general form is an **interval with its coverage** (68% or 95%): no assumed distribution; may be asymmetric. $\pm$ is a symmetric interval; read as one standard uncertainty, it covers 68% only for Gaussian errors. State N and the method (here: bootstrap of the 9 timings, Part B).

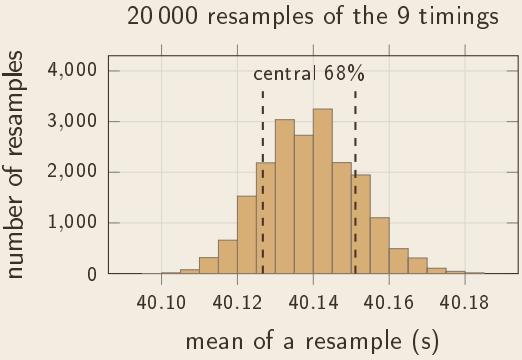
PART B

Propagating uncertainty

From the uncertainty of what we measure to the uncertainty of what we compute.

Resampling the data: the bootstrap

Problem: the uncertainty of a statistic computed from the 9 timings – mean, median, anything – without assuming a distribution.

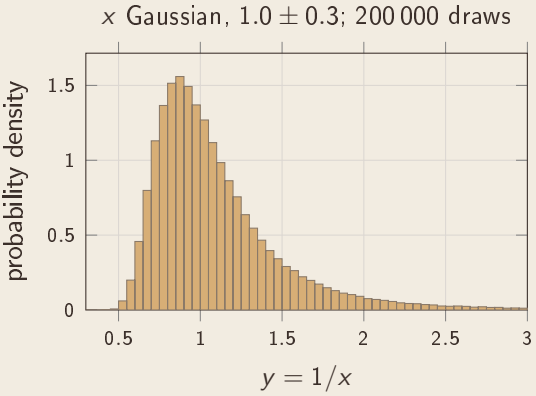


1. Draw 9 of the 9 values **with replacement**.
2. Compute the statistic.
3. Repeat; the spread (standard deviation) of the results is the uncertainty; their central 68% (16th to 84th percentile) is the interval.

	value	spread	68% interval
mean	40.139	0.012	40.127–40.151
median	40.130	0.019	40.120–40.150

Mean: agrees with $s/\sqrt{N} = 0.013$ s (slide 10).
Assumes independent values.

Propagating through any function: Monte Carlo



Problem: $y = f(x)$ with x uncertain; how uncertain is y? Example: $y = 1/x$.

Monte Carlo: draw many x, compute y for each, describe the y values: median 1.00, central 68%: 0.77 to 1.43 – **asymmetric**. (A mean of y does not exist here: draws near $x = 0$ make y unbounded.)

Where the x values come from

Measured repeatedly: bootstrap resamples of the data (nonparametric).
Given once (a calibration certificate, a tolerance): the stated distribution. Here: x Gaussian, 1.0 ± 0.3.

A shortcut when f is nearly straight: follow the slope

Problem: a formula instead of a simulation. $y = f(x)$, with $x = \bar{x} \pm \sigma_x$ (σ_x: standard uncertainty of x, e.g. $s/\sqrt{N}$). If f is nearly straight over $\bar{x} \pm \sigma_x$:

$$\boxed{\sigma_y \approx |f'(\bar{x})|\,\sigma_x}$$

Function	Result	Example
$y = ax$	$\sigma_y = \lvert a\rvert\,\sigma_x$	period $T = \bar{t}_{20}/20$, with $\bar{t}_{20} = 40.139 \pm 0.013\,\mathrm{s}$ the mean time of 20 swings: $\sigma_T = 0.0007\,\mathrm{s}$
$y = x^n$	$\sigma_y/\lvert y\rvert = \lvert n\rvert\,\sigma_x/\lvert x\rvert$	$1/T^2$: relative uncertainty doubled

Relative uncertainties multiply by the power: a quantity that enters squared counts twice.
Check against Monte Carlo: for $1/x$ at $x = 1.0 \pm 0.3$ the rule gives a symmetric ± 0.3 and misses the tail of the previous slide; at $x = 1.00 \pm 0.05$ both give 0.95–1.05. The rule assumes no distribution; reading $\pm\sigma_y$ as 68% does.

Several independent inputs: the pendulum

For $y = f(x_1, \ldots, x_n)$ with **independent** inputs x_i of standard uncertainties σ_i:

$$\sigma_y^2 = \sum_i \left(\frac{\partial f}{\partial x_i}\right)^2 \sigma_i^2 \qquad \text{for } g = \frac{4\pi^2 L}{T^2}: \quad \left(\frac{\sigma_g}{g}\right)^2 = \left(\frac{\sigma_L}{L}\right)^2 + \left(\frac{2\sigma_T}{T}\right)^2$$

Input	Value	Contribution to σ_g/g
L	1.000 ± 0.002 m	0.20%
T	2.0069 ± 0.0007 s	$2 \times 0.033\% = 0.07\%$
Total (in quadrature)		0.21%

Result

$g = 9.80 \pm 0.02\ \mathrm{m\,s^{-2}}$

Monte Carlo check (T: bootstrap; L: Gaussian ± 0.002 m, its tolerance): spread 0.020; 68%: 9.78–9.82.

To improve it, reduce the **largest** contribution first: here L.

Correlated inputs: do not ignore the correlation

With correlated inputs a cross term appears; for two inputs a, b with correlation coefficient ρ (-1 to 1; Lecture 3):

$$\sigma_y^2 = \left(\frac{\partial f}{\partial a}\right)^2 \sigma_a^2 + \left(\frac{\partial f}{\partial b}\right)^2 \sigma_b^2 + 2\,\frac{\partial f}{\partial a}\frac{\partial f}{\partial b}\,\rho\,\sigma_a\sigma_b$$

Straight line of Lecture 3: $y = a + bx$, $a = 1.95 \pm 0.18$, $b = 0.499 \pm 0.033$, $\rho = -0.84$

x	y	σ_y with ρ	σ_y ignoring ρ
0	1.95	0.18	0.18
4.5 (centre of data)	4.20	0.10	0.23
10 (beyond the data)	6.94	0.20	0.38

Ignoring the correlation here overstates σ_y by up to 2.4 times; with $\rho > 0$ it would understate it.

Random and systematic uncertainties

	Random (independent)	**Systematic (common)**
Example	stopwatch reaction time	a ruler that is 0.1% too long
More data	reduces it as $1/\sqrt{N}$	does not reduce it
Summed over N uses	in quadrature: $\sqrt{N}\times$	linearly: $N\times$
Detected by	scatter, χ^2, pulls	only independent checks

Ten lengths of 1.000 m, laid end to end

Independent errors of ± 0.002 m each: total ± 0.006 m ($\sqrt{10} \times 0.002$). A ruler 0.1% long: every length $+0.001$ m, total $+0.010$ m.

PART C

Testing hypotheses

Is a difference real, or could chance have produced it?

The p-value

1. State a **null hypothesis** H_0: e.g. "the two groups of timings come from the same distribution".
2. Compute a **test statistic** from the data, e.g. the difference of the two groups' medians.
3. The **p-value**: the probability, *if H_0 is true*, of a statistic at least as extreme (as far from what H_0 predicts) as the one observed.

Reading it

Small p: the data would be surprising if H_0 were true. Large p: the data are compatible with H_0 – which does **not** show that H_0 is true.

Two families of tests

Step 3 needs a **reference distribution**: how the statistic would be distributed if H_0 were true.

	Nonparametric	Parametric
Reference distribution	built from the data themselves: by reshuffling, or from ranks	derived from an assumed distribution of the data
Examples	permutation tests, rank tests, shuffle surrogates	z, Student's t, χ^2
Assumes	independence only	independence and the distribution
Valid when	any shape	the assumed distribution is right (or N is large)

Common to both

The definition of p is the same; only the way its reference distribution is obtained differs. Neither family protects against dependent values or systematic errors.

Do two groups differ? Permutation and rank tests

Problem: do groups A and B differ? H_0: same distribution.
A: the 9 timings of slide 6 + the bad value 41.14 s.
B (second session): 40.17, 40.26, 40.20, 40.23, 40.19, 40.28, 40.22, 40.24, 40.18, 40.21 s.

Permutation: under H_0 the labels do not matter. Shuffle them many times, recompute the difference of medians; p = fraction at least as large as observed.

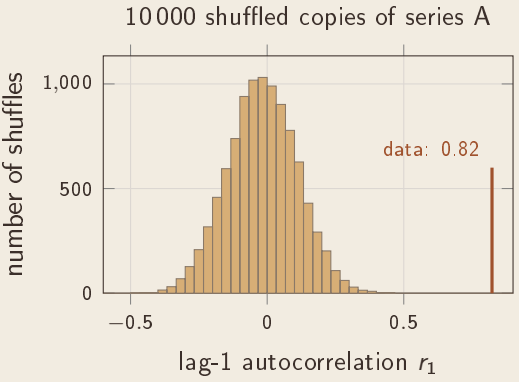
Rank (Mann–Whitney): rank all 20 values; are A's ranks unusually low or high?

p (two-sided)	with bad value	without
permutation	0.005	0.002
rank	0.011	0.002
t on means*	0.84	0.0003

*parametric: assumes Gaussian timings (slide 24)

Permutation and rank tests see the difference with or without the bad value; the t-test loses it.
Effect: B longer by 0.07 s, 95% interval 0.03–0.11 s (Hodges–Lehmann: median of all 10×10 differences B−A), bad value included.

Are successive values independent? Shuffle test



H_0: the values are **independent** – their order does not matter. Shuffling keeps the values (same histogram) and destroys the order.

$r_1 = \dfrac{\sum_i (x_i - \bar{x})(x_{i+1} - \bar{x})}{\sum_i (x_i - \bar{x})^2}$: how similar neighbours are.

Series A of Lecture 2 ("Which one is random?": a sine plus noise, 60 values): $r_1 = 0.82$. Shuffled copies: largest 0.46; none of 10 000 reaches 0.82, so $p < 10^{-4}$.

Surrogate data

Copies made to obey H_0 while keeping everything else. Shuffling tests only independence.

Parametric shortcuts: z and t

Problem: with only quoted values, or a distribution known from outside the data, a formula gives the reference.

Two quoted values: z

$g = 9.80 \pm 0.02$ (ours) and $9.92 \pm 0.06\,\mathrm{m\,s^{-2}}$: only summaries, nothing to reshuffle.

$z =$ difference/its uncertainty $= 0.12/0.063 = 1.90$.

If the errors are Gaussian: two-sided $p = 0.058$. **Not conclusive**: report 0.12 ± 0.06.

Few values against a reference: t

First 8 timings of slide 6 vs. calibration $\mu_0 = 40.20\,\mathrm{s}$. H_0: centred on μ_0.

Sign flip: under H_0 each difference from μ_0 is equally likely $\pm$; all $2^8 = 256$ patterns: $p = 0.016$.

$t = (\bar{x} - \mu_0)/(s/\sqrt{N}) = -4.05$; if Gaussian, Student's t distribution ($\nu = 7$): $p = 0.005$.

The smaller p of the t-test comes from the assumed Gaussian shape, which 8 values cannot check.

Parametric: goodness of fit as a test

If the model and the σ_i are right **and the errors are independent and Gaussian**, χ^2 follows the χ^2 distribution with $\nu = N -$ (number of fitted parameters) degrees of freedom, and

$$p = P\left(\chi^2_\nu \geq \chi^2_{\text{observed}}\right).$$

Fits from Lecture 3	χ^2	ν	p
Line on straight data	4.2	8	0.84
Quadratic on curved data	13.7	18	0.75
Line on curved data	90	19	3×10^{-11}

H_0: "the model describes the data within the uncertainties". A tiny p rejects it – but look at the residuals to see *why*. Without assuming Gaussian errors: a shuffle test of the residuals (slide 23) checks whether they are independent.

What a p-value is not

p **is not . . .**	**because . . .**
the probability that H_0 is true	it is computed *assuming* H_0 is true
the probability that the result is a fluke	the same statement in other words
the size or importance of an effect	a tiny effect gives a tiny p with enough data
a sharp line at 0.05	$p = 0.045$ and $p = 0.055$ are the same evidence

Report the effect with its interval (or $\pm$); the p-value only adds to it.

Two kinds of error

	H_0 **true**	H_0 **false**
reject H_0	false alarm (probability α)	correct (power $1-\beta$)
keep H_0	correct	miss (probability β)

The threshold α is chosen **before** looking at the data. It sets the false-alarm rate:

Convention	threshold	false alarms
many fields	$p < 0.05$	1 in 20
particle physics, discovery	$p < 3 \times 10^{-7}$ ("5σ", one-sided)	1 in 3.5 million

Many tests: false alarms add up

20 independent tests, no real effect anywhere, threshold $\alpha = 0.05$:

$$P(\text{at least one "significant" result}) = 1 - 0.95^{20} = 0.64.$$

Remedies

Decide the tests before looking. With m tests, use α/m (Bonferroni: 0.0025 for 20). Report how many tests were done.

Where it hides

Trying several models, cuts, time windows or variables, and reporting the one that "worked".

The assumptions behind every test

Assumption	**If it fails**
Values are independent	standard errors and p-values usually too small – common for time series
The distribution is right (parametric tests)	use a nonparametric test: permutations, ranks, surrogates
The test was chosen before looking	p loses its meaning (hidden multiple testing)
No systematic errors	p says nothing about them

A p-value is only as good as its assumptions.

Summary

Task	Tools	Check
Describe	plot in order; median, MAD, quantiles; $\bar{x}$, s; a distribution family only when known	independence; bad values
Propagate	bootstrap; Monte Carlo; the linear rule when f is nearly straight	correlations; curvature of f; systematic errors
Test	permutations, ranks, surrogates; z, t, χ^2 when the distribution is known	assumptions; number of tests; report the effect with its interval

Before you report a result: five questions

1. Did I plot the values in the order they were taken? *(drift, time variability)*
2. Are the values independent? *(otherwise uncertainties and p-values are too small)*
3. Which uncertainties are random, and which systematic? *(quadrature or linear)*
4. Does my method assume a distribution or a straight-line approximation, and do I know that independently of these data? *(if not: resampling, permutations, ranks, Monte Carlo)*
5. What exactly is H_0, and did I choose the test before looking? *(multiple testing)*

Then quote: value, interval (or $\pm$), units, and how they were obtained.

TO TAKE AWAY

A number without an uncertainty is not a result.

And a p-value without its assumptions is not evidence. Plot in order, check independence, propagate honestly, then test.